

GALO REYES

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Computer science student with hands-on systems programming experience in C — kernel modules, file systems, and multithreading — seeking a software engineering internship in health/dental technology.

EDUCATION

San Francisco State University

Expected Grad: May 2027

B.S. Computer Science | GPA: 3.35

- Relevant coursework: Operating Systems, Data Structures, Algorithms, Database Systems, Software Engineering

PROJECTS

Custom Unix-Style File System (4-person team)

C, Linux | Summer 2026

- Built a working file system from scratch on a raw block device: volume formatting, free-space management, hierarchical directories, and file read/write operations.
- Implemented a buffered I/O layer (`b_open/b_read/b_close`) that serves byte-level reads of any size on top of fixed 512-byte block operations, using per-file control blocks to support multiple concurrently open files.
- Collaborated in a 4-person team using Git feature branches and code review on a shared repository.

Linux Kernel Device Driver — `csc415crypt`

C, Linux Kernel | Summer 2026

- Wrote a loadable character device driver that performs reversible key-based encryption on byte streams via `/dev/csc415crypt`, with dynamic major-number allocation and automatic device-node creation.
- Implemented full file operations (`open`, `read`, `write`, `release`, `lseek`) plus an `ioctl` control channel for setting keys, switching encrypt/decrypt modes, and clearing state; used `copy_from_user/copy_to_user` for safe kernel-user data transfer.
- Protected all shared driver state with a mutex for safe concurrent access; debugged a failed decrypt round-trip to a missing `.lseek` handler and verified the fix through kernel logs.

Multithreaded Police Dispatch Data Analyzer

C, pthreads | Summer 2026

- Built a concurrent statistics engine over a large fixed-length-record dataset of police dispatch calls, computing per-call-type response-time distributions (median, quartiles, mean, std dev) across configurable thread counts.
- Used a hash table keyed by call type with a mutex per bucket instead of one global lock, so threads updating different call types never block each other; benchmarked 1/2/4/8-thread runs to show scaling limited by the VM's core count.

SwampMeet Marketplace (Team) + Portfolio Website

HTML/CSS/JS, MySQL | 2026

- Built SwampMeet with a CSC 648 team: a marketplace web app for SFSU students with `@sfsu.edu`-only accounts, course-code textbook matching, and campus-safe meetup zones — live at `swampmeet.vercel.app`.
- Designed and built a responsive portfolio site from scratch with mobile navigation and scroll-reveal animations; deployed with a QR code for in-person networking.
- Built a Java application backed by MySQL using the DAO pattern — designed the relational schema, seeded sample data with SQL scripts, and organized code into data-access, utility, and test layers.

TECHNICAL SKILLS

Languages: C, C++, Java, SQL (MySQL), JavaScript, HTML/CSS

Systems: Linux kernel modules, POSIX threads, file systems, low-level I/O, memory management, gdb, make

Tools: Git/GitHub, Linux command line, VS Code, virtual machines

EXPERIENCE & ACTIVITIES

Image Orthodontics

Dental Assistant | 2024 - Present

- Designed and built a patient self-scheduling web demo based on workflow gaps observed on the job: online booking with insurance intake, an AI assistant for common questions, QR-code booking, and an SMS fallback for dropped chats.
- Proposed recall/reminder workflow improvements (6-month recall texts with self-booking links) to reduce phone-tag and no-shows for a practice running CareStack and Cloud 9.
- Attending Bay Area Dental Expo 2026 to pursue software internships in dental technology (imaging, AI diagnostics, practice management).